io_uring C++

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1

1:

	ServerCore — io_uring C++ + C++ DirectX 11 — FactionClash Binary Header `[size:uint16][msgId:uint16]` + Protobuf 3 payload

1.1 io_uring

Linux , epoll Linux I/O `io_uring` .

epoll `read/write` , I/O - . `io_uring` Submission Queue Completion Queue , Multishot Accept/Recv SQE .
I/O , .

1.2

ServerCore (`IoRing`, `Session`, `BufferPool`) (`IoLoop`, `GameRoom`, `PacketSession`)
. `PlayerId`, `RoomId` , HTTP DB .

. `IoLoop`, `GameRoom`, `PacketSession`, `IoCommand`
. servercore I/O, , , , .

1.3

2 io_uring

ServerCore Linux `io_uring` . Submission Queue(SQ) Completion Queue(CQ)
, I/O .

`IoRing` `io_uring` RAII , . .

- **Multishot Accept** — SQE . CQE SQE accept .
- **Multishot Recv + Provided Buffer** — `IOSQE_BUFFER_SELECT` Provided Buffer Ring
, . .
- **MSG_RING** — `IORING_OP_MSG_RING` .

2.1 Dispatch() — CQE

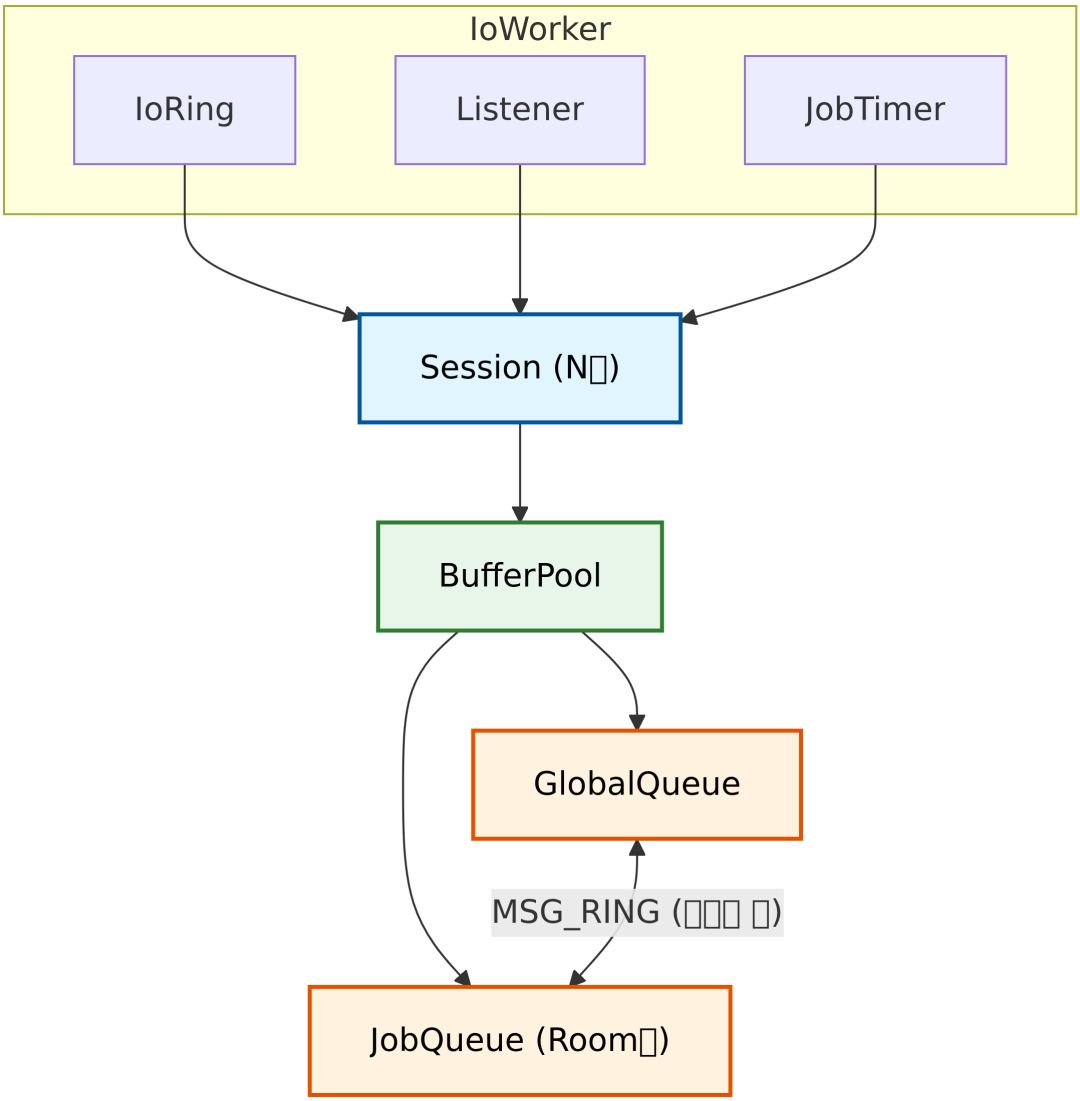
`Dispatch()` CQ dispatch . `user_data` bit 0 MSG_RING I/O .

MSG_RING `move_only_function` bit 0 `user_data` . I/O `IoEvent*`
, `keep_alive` .

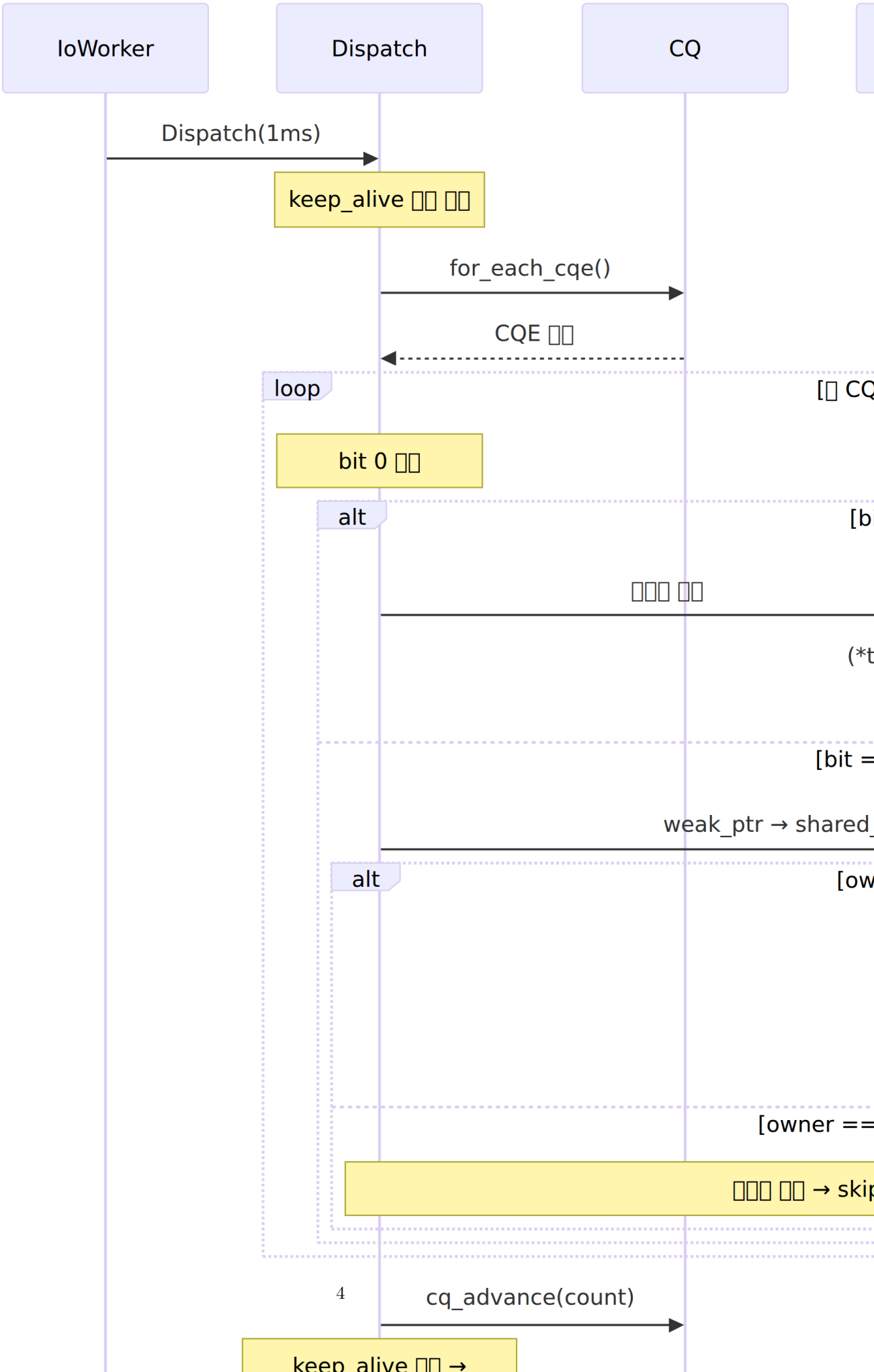
2.2 Provided Buffer Ring

Provided Buffer Ring `mmap` (`io_uring_buf`) .

- `RingBuffer::Create()` `meta_size + data_size` `mmap` TLB .
- `io_uring_buf_ring_add` .
- recv , .



1: ServerCore v4


IoWorker
Dispatch
CQ
Dispatch(1ms)
keep_alive □□ □□
for_each_cqe()
CQE □□
loop
[□ CQ
bit 0 □□
alt
[bi
□□□ □□
(*t
[bit =
weak_ptr → shared_
alt
[ow
[owner ==
□□□ □□ → skip
cq_advance(count)
keep_alive □□ →

1 IoRing.cpp --- Dispatch(): CQE

```
bool IoRing::Dispatch(std::chrono::milliseconds timeout) {
    std::vector<EventHandlerRef> keep_alive;
    unsigned head = 0; unsigned count = 0;
    io_uring_for_each_cqe(ring_, head, cqe) {
        const std::uint64_t data = cqe->user_data;
        if (data & 0x1ULL) {  // MSG_RING    (bit 0   )
            auto* task = reinterpret_cast<std::move_only_function<void()>*>(
                data & ~0x1ULL);
            (*task)(); delete task;
        } else if (data != 0) {  //   I/O
            auto* ev = reinterpret_cast<IoEvent*>(data);
            auto owner = ev->Owner();
            if (owner) {
                keep_alive.push_back(owner);  //
                owner->Dispatch(ev, result, flags);
            }
        }
        ++count;
    }
    io_uring_cq_advance(ring_, count);
    return true;
}
```

2.3

Provided Buffer Ring .

(ENOBUFS) . provided buffer `ENOBUFS` Multishot Recv . , CQE `RegisterRecv()` Multishot Recv .

. TCP CQE , provided buffer CQE zero-copy . .

`Session::OnRecv()` Fast Path / Slow Path . CQE (Fast Path) `F_MORE` `--pending_io_` , `span` `Return()` . (Slow Path) `PacketSession` `RecvBuffer` , `ENOBUFS` `RegisterRecv()` Multishot Recv . , .

3

3.1 self_ref_ + pending_io_

. SEGFAULT . CQE `Session` , .

I/O . `Session` , SQE(Multishot Recv, Send) . SQE CQE **Use-After-Free** . , .

I/O . `Session` .

- `self_ref_` — `shared_ptr` . `self_ref_` .
- `pending_io_` — CQE SQE . SQE `++pending_io_`, CQE `--pending_io_` .

. `TryRelease()` `pending_io_ > 0` , `pending_io_` 0 — in-flight I/O — `OnDisconnected()` `self_ref_.reset()` . , UAF .

`pending_io_` 0 `Disconnect()` cancel `shutdown(SHUT_RDWR)` SQE . shutdown Multishot Recv EOF(`res == 0`) CQE , in-flight Send CQE , SQE

2 `include/servercore/io/Session.h`

```
// Counts in-flight io_uring ops whose CQEs reference Session members.
// self_ref_ must not be released until this reaches zero.
// Accessed from I/O thread only - no synchronization needed.
int pending_io_ = 0;

// Self-ownership: keeps Session alive during active I/O.
// Released only via TryRelease() after all in-flight ops complete.
std::shared_ptr<Session> self_ref_;
```

.

3.2 CQE

`pending_io_` . `Dispatch()` CQE , . `TryRelease()` , UAF .

+1 . CQE `IoEvent::Owner()` `weak_ptr` `shared_ptr` `keep_alive` , `self_ref_.reset()` . `keep_alive` , .

3.3 IoEvent weak_ptr

`pending_io_` `keep_alive` , . Heartbeat I/O , `IoEvent` `shared_ptr` in-flight I/O .

`IoEvent` `weak_ptr` . CQE `Owner()` `weak_ptr` → `shared_ptr` `nullptr` . , CQE .

4

4.1 Ring-per-thread

io_uring ring-per-thread . Multishot Accept Accept , Recv/Send . . accept , Room Match .

`SO_REUSEPORT` . `IoWorker` `Listener` `accept()` , .

`IORING_OP_MSG_RING` opcode . CQ fd , Room .

4.2 SO_REUSEPORT — OS

`IoWorker::Start()` `SO_REUSEPORT` `Listener` , .

4.3 MSG_RING —

`RunOnRing()` bit 0 . , `IORING_OP_MSG_RING` SQE . `IOSQE_CQE_SKIP_SUCCESS` CQE .

5 — IoWorker

`IoWorker` I/O . , 8ms Room I/O .

2: IoWorker

	/
Phase 1 I/O	`ring_->Dispatch()` — CQE
Phase 2	`ProcessPostedTasks()` —

	/
Phase 3	`timer_.DistributeExpired()` —
Phase 4	`GlobalQueue` `JobQueue` 8ms

5.1 JobQueue &

Room . Room `IoWorker` , Room Room . `JobQueue` + `GlobalQueue` .

5.1.1 JobQueue — -

`JobQueue` Room · Match . `Push(job)` , .

:

1. **Batch Swap** — , `batch` . , Push Execute .
2. — `Execute(deadline)` `steady_clock::now()` deadline(8ms) . 8ms 60fps (16.6ms) , I/O .
3. — , `JobQueue` `GlobalQueue` . Phase 4 .

5.1.2 GlobalQueue —

`GlobalQueue` `JobQueue` . `IoWorker` Phase 4 `JobQueue` .

- `IoWorker` `GlobalQueue` `JobQueue` , Room .
- `JobQueue` .

5.1.3 JobTimer —

`JobTimer` `steady_clock` . Room (20Hz) .

- — `Reserve(delay, callback)` .
- — `IoWorker` Phase 3 `DistributeExpired()` , `JobQueue` Push . `JobQueue` , .
- — `Reserve` .

6 PostgreSQL — DB

PostgreSQL `IoWorker` `libpq` , Room . io_uring DB .

`PQsendQuery` + `POLL_ADD` PostgreSQL fd readiness io_uring . . `POLL_ADD` readiness , `recv(2)` `PQconsumeInput()` libpq . io_uring Provided Buffer Ring Multishot Recv DB , `epoll` + libpq .

`libpq` .

(`Connecting/QuerySent/ReadingResult`) `POLL_ADD` `PQexec` , DB `IoWorker` . sqlpp11 ORM .

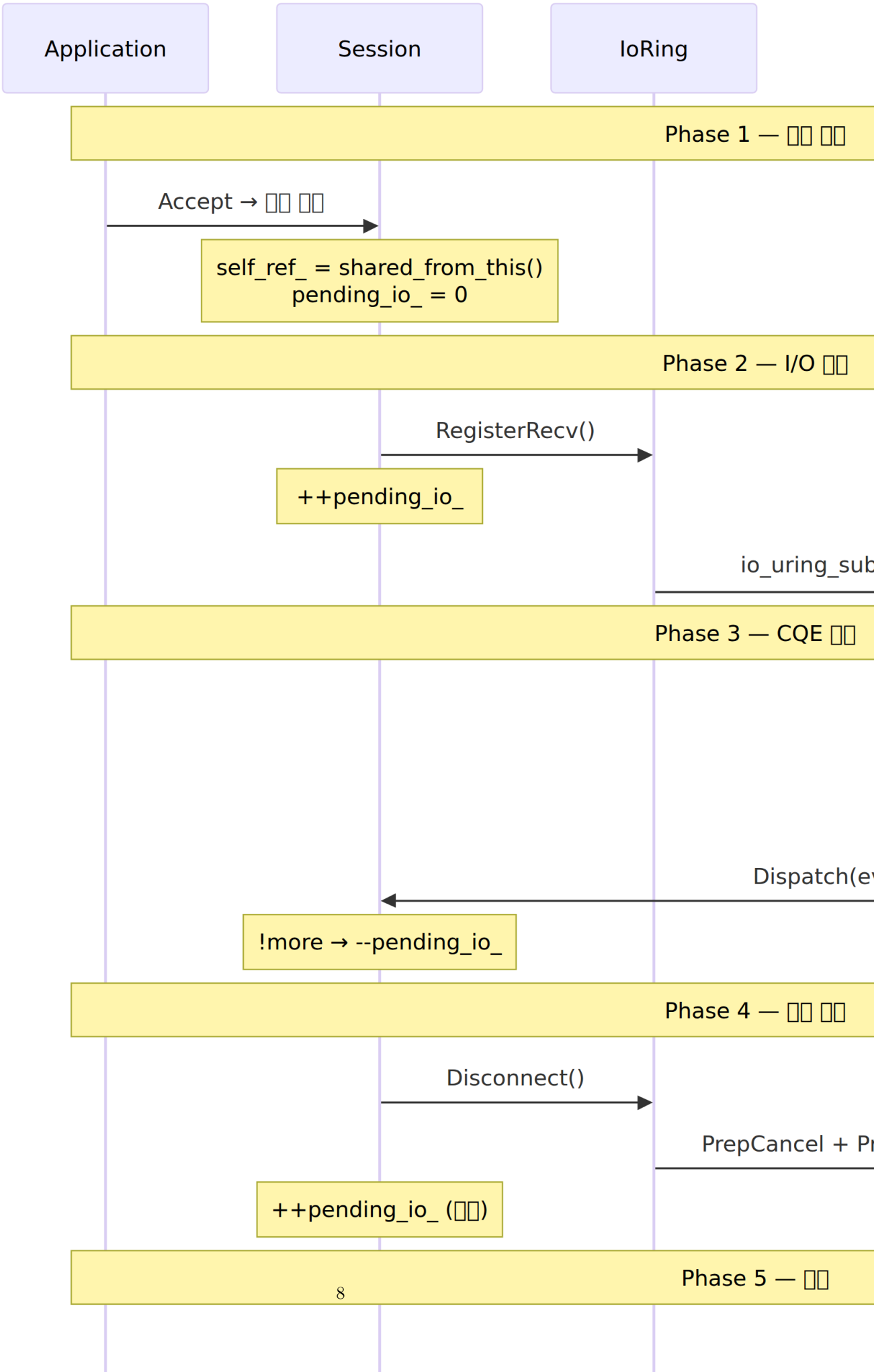
DB

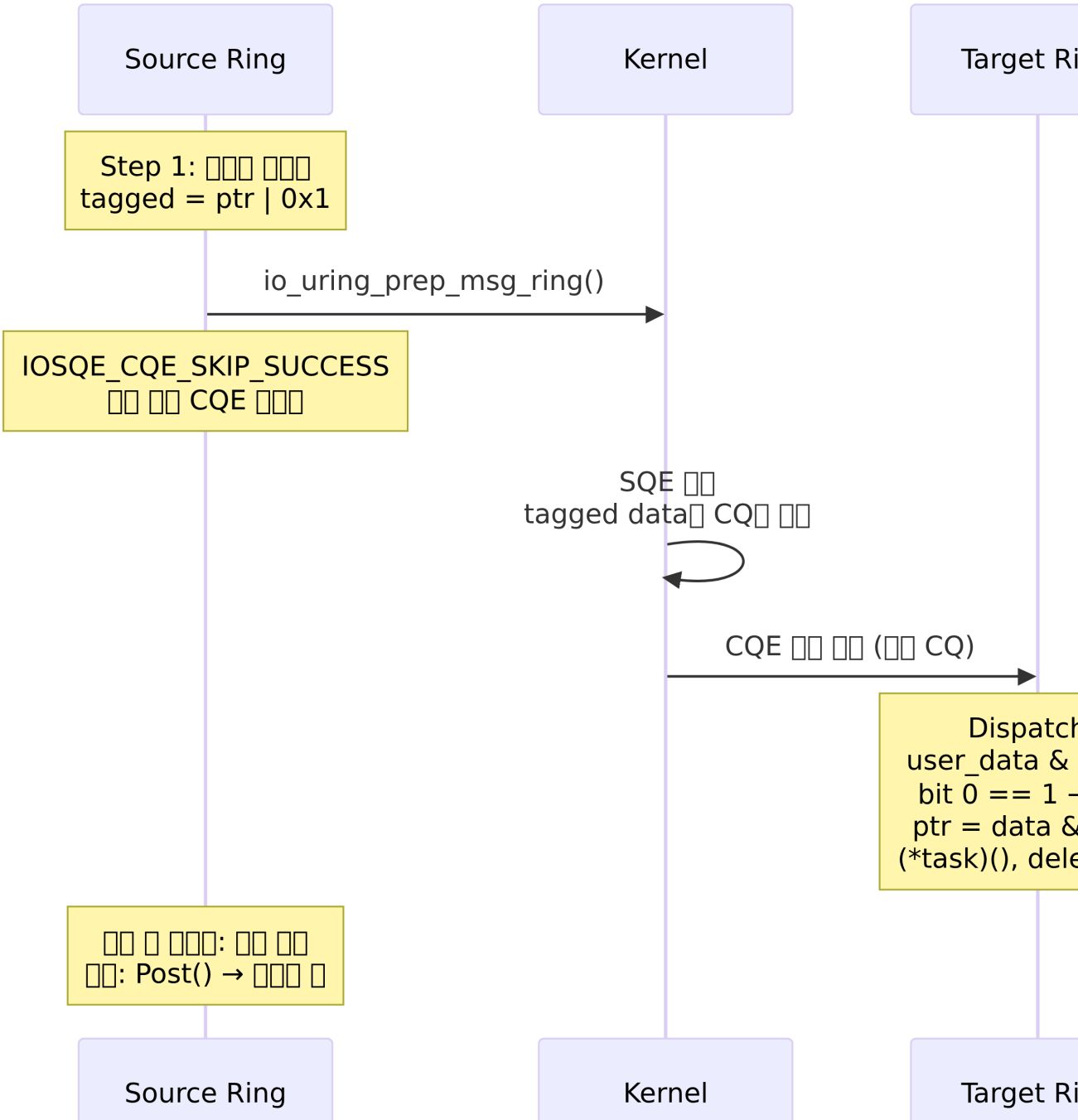
DB . DB Room , `IoWorker` .

`IoRing` `thread_local IoRing* Current()` . `PgConnectionPool::Execute()` , DB `PQexec` `Post(lambda)` . `IoWorker` `ProcessPostedTasks()` , Room .

`PgConnectionPool` — - 1:1

`PgConnectionPool` facade , `PgWorkerPool` . `PgWorkerPool` `connections_per_worker` , **`PgConnection` 1:1** . , `std::mutex` + `std::condition_variable` . `PQexec` .


Application
Session
IoRing
Phase 1 — □□ □□
Accept → □□ □□
self_ref_ = shared_from_this()
pending_io_ = 0
Phase 2 — I/O □□
RegisterRecv()
++pending_io_
io_uring_sub
Phase 3 — CQE □□
Dispatch(ev
!more → --pending_io_
Phase 4 — □□ □□
Disconnect()
PrepCancel + Pr
++pending_io_ (□□)
Phase 5 — □□


Source Ring
Kernel
Target Ri
Step 1: □□□ □□□
tagged = ptr | 0x1
io_uring_prep_msg_ring()
IOSQE_CQE_SKIP_SUCCESS
□□ □□ CQE □□□
SQE □□
tagged data□ CQ□ □□
CQE □□ □□ (□□ CQ)
Dispatch
user_data &
bit 0 == 1 –
ptr = data &
(*task)(), dele
□□ □ □□□: □□ □□
□□: Post() → □□□ □
Source Ring
Kernel
Target Ri

4: MSG_RING

3 IoRing.cpp --- RunOnRing(): MSG_RING

```
bool IoRing::RunOnRing(std::move_only_function<void()> task) noexcept {
    if (t_current_ == this) { task(); return true; }  //
    if (t_current_ && t_current_->ring_) {
        auto* task_ptr = new std::move_only_function<void()>(std::move(task));
        std::uint64_t tagged = reinterpret_cast<std::uint64_t>(task_ptr) | 0x1ULL;
        io_uring_sqe* sqe = io_uring_get_sqe(t_current_->ring_);
        io_uring_prep_msg_ring(sqe, Fd(), 0, tagged, 0);
        sqe->flags |= IOSQE_CQE_SKIP_SUCCESS;  //     CQE
        io_uring_submit(t_current_->ring_);
        return true;
    }
    Post(std::move(task)); return true;
}
```

4 io_worker.cpp --- Run(): IO +

```
void IoWorker::Run() {
    IoRing::SetCurrent(ring_.get());
    while (running_) {
        ring_->Dispatch(std::chrono::milliseconds(1));     // 1. I/O
        ring_->ProcessPostedTasks();                        // 2.
        timer_.DistributeExpired();                         // 3.
        while (auto* q = global_queue_.TryPop())
            q->Execute(std::chrono::steady_clock::now()
                       + std::chrono::milliseconds(8));     // 4.     (8ms  )
    }
}
```

.

- — PQconnectdb . TCP .
- — . PostgreSQL .
- — . DB .

3:

1000	1000	→ DB	(: 20) ,
5000	5000	/	20

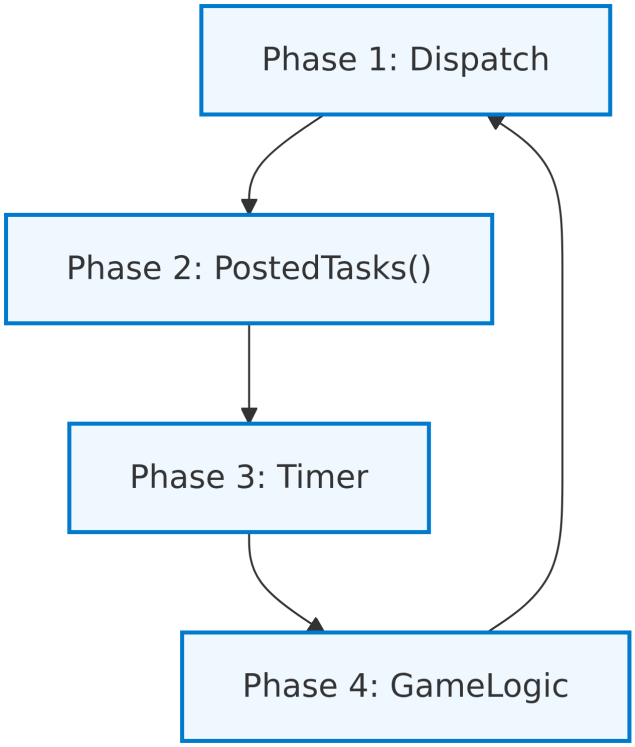
7

, , , DB . , .

7.1

[size:uint16][msgId:uint16] + Protobuf 3 payload . PacketBuilder , msgId . 20Hz(kTickInterval = 50ms) , · Room .

7.2



5: IoWorker

4:

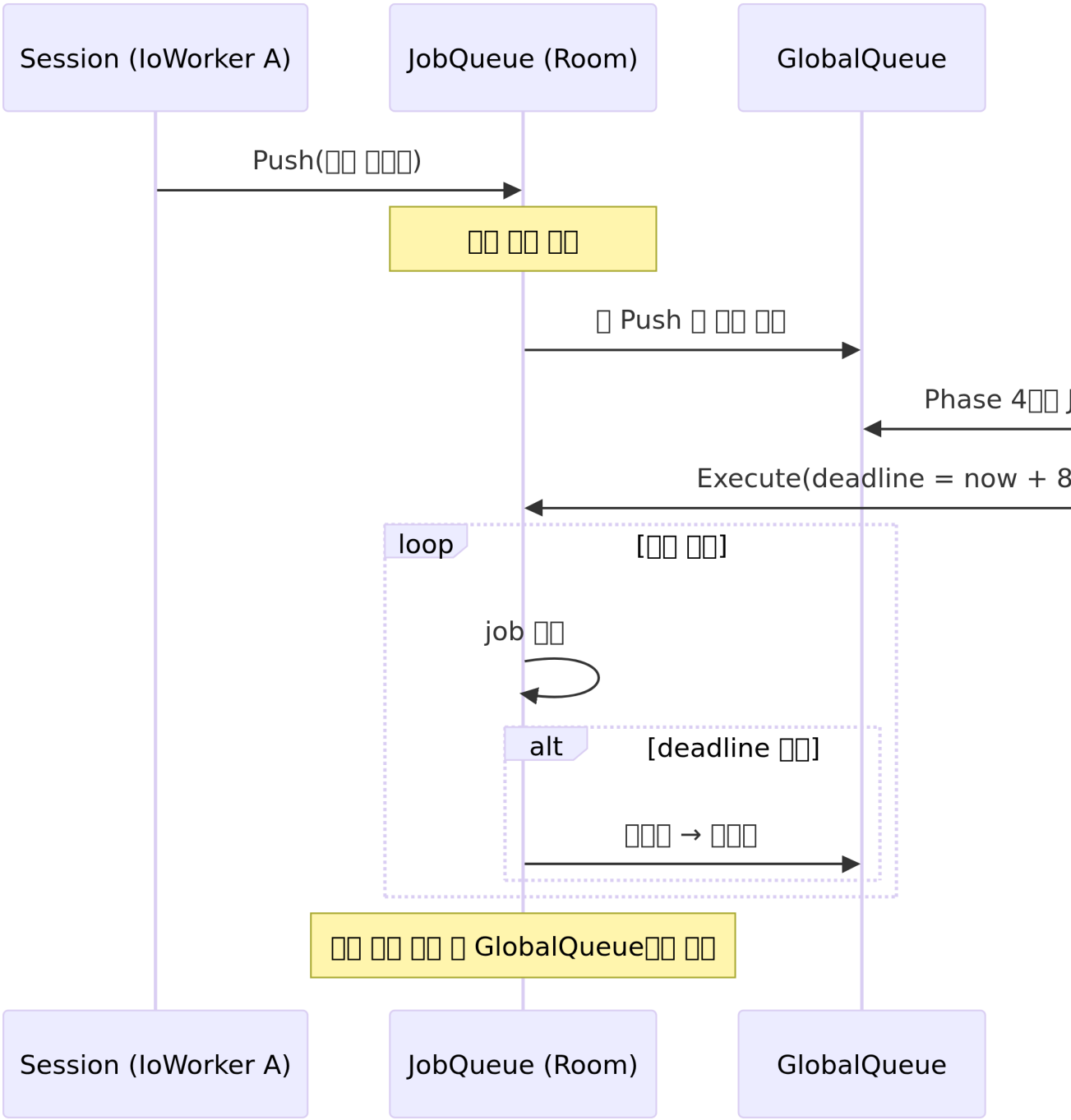
.	4 (Basic Attack / Power Strike / Fireball / Heal), · , `S_DAMAGE` All / Zone / Party / Whisper 4 , Room / / `PartySystem` , `ItemInstance` , 3 (Gold/Gems/Tokens), `DbService`

. `C_ATTACK` / `C_FIRE` , , . `CombatSystem::kSkills` , `skill_cooldowns` . . / , Zone All Room . `DbService` .

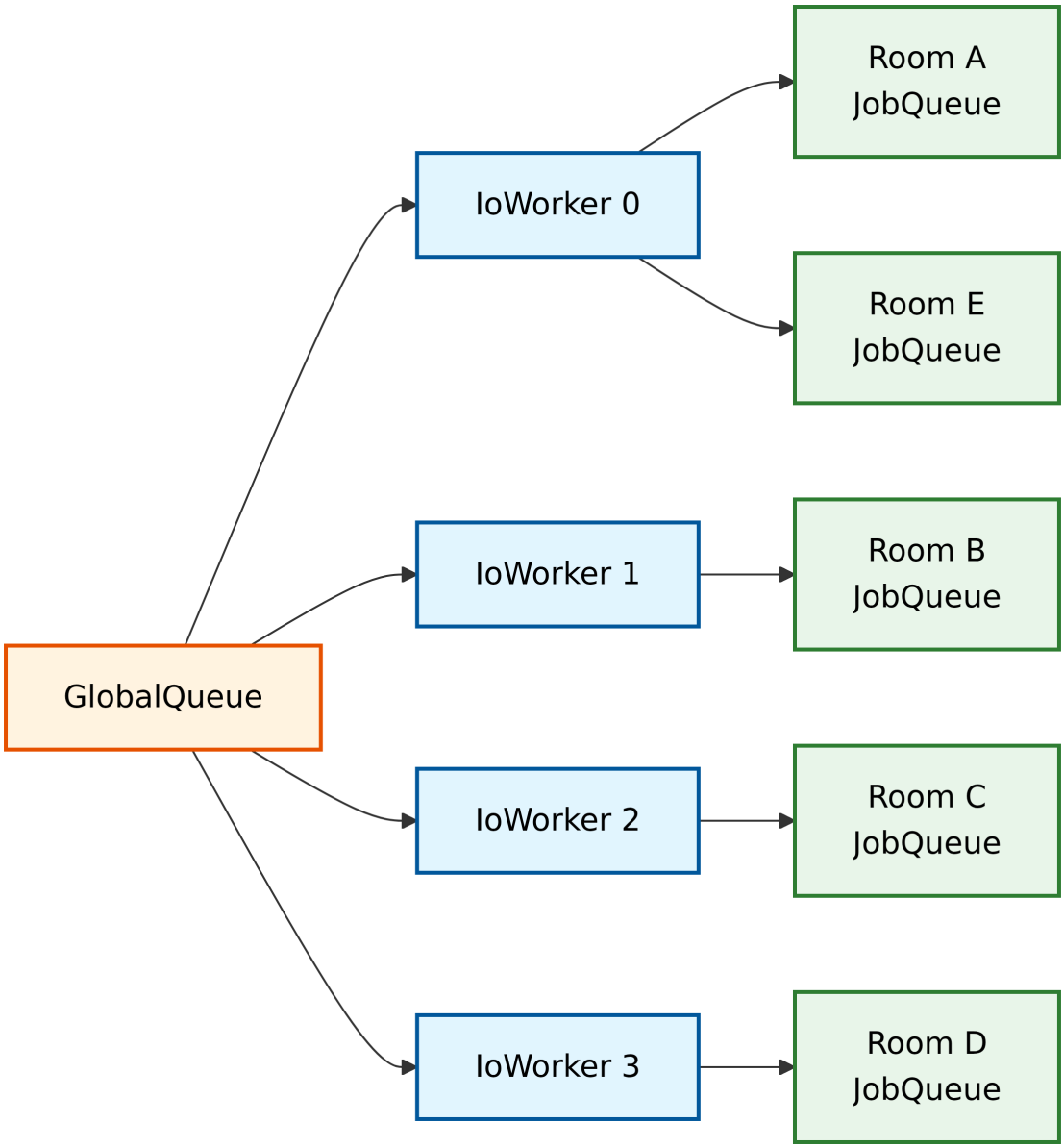
7.3

120×70 Cellular Automata , .

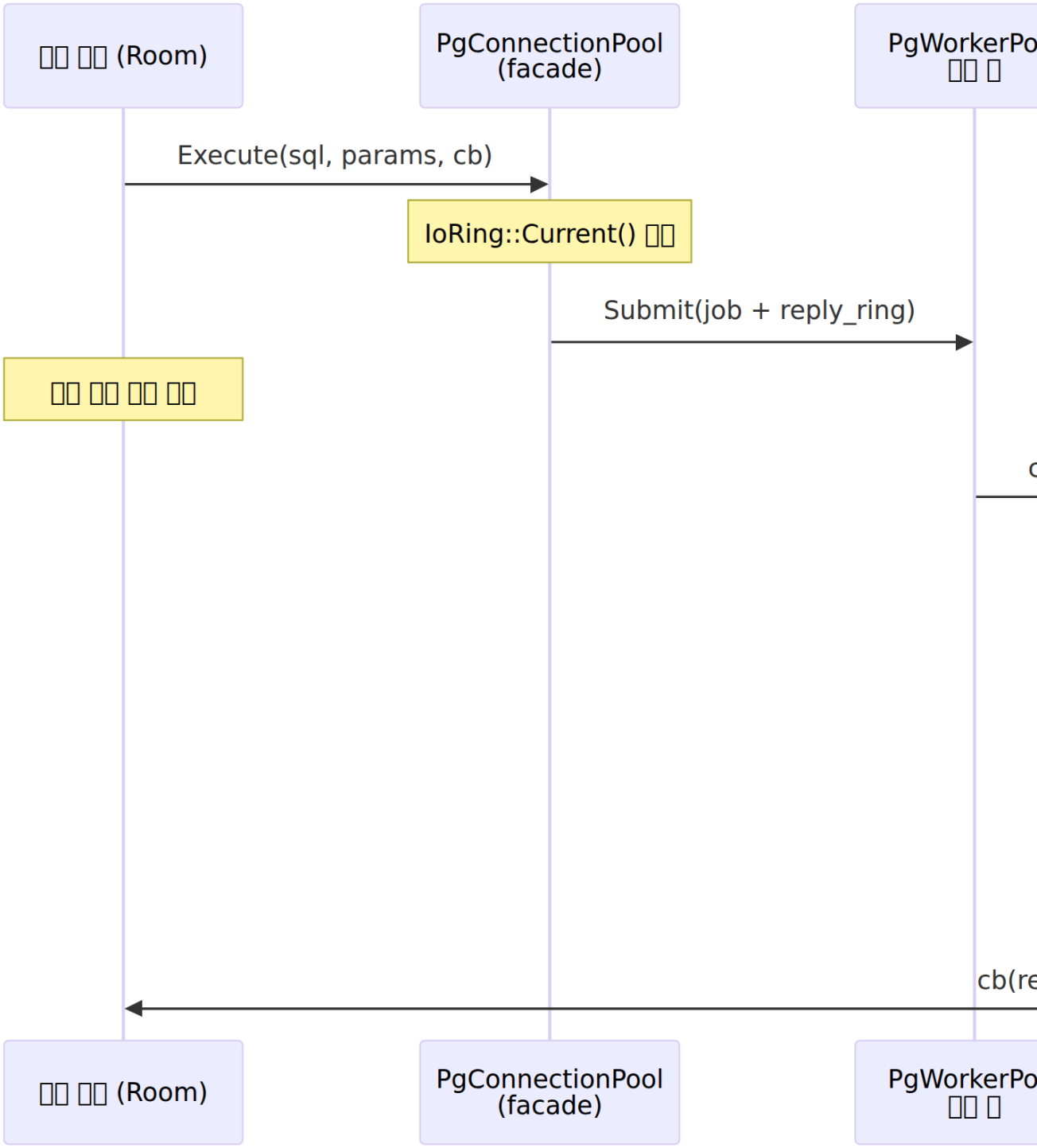
1. **RandomFillMap** — 52% (Zone depth +2% , 62%). Zone .
2. **SmoothMap ×5** — 8 .
3. **ProcessMap** — Flood Fill , 15 20 . .
4. **ConnectClosestRooms** — edge tile 1 . Main Room .
5. **PlacePortals** — 5×5 15 . 8% 3 .
6. **FindSpawnPosition** — Main Room 5×5 ( ) .


Session (IoWorker A)
JobQueue (Room)
GlobalQueue
Push(□□ □□□)
□□ □□ □□
□ Push □ □□ □□
Phase 4□□ J
Execute(deadline = now + 8
loop
[□□ □□]
job □□
alt
[deadline □□]
□□□ → □□□
□□ □□ □□ □ GlobalQueue□□ □□
Session (IoWorker A)
JobQueue (Room)
GlobalQueue

6: JobQueue



7: GlobalQueue


□□ □□ (Room)
PgConnectionPool
(facade)
PgWorkerPo
□□ □
Execute(sql, params, cb)
IoRing::Current() □□
Submit(job + reply_ring)
□□ □□ □□ □□
c
cb(re
□□ □□ (Room)
PgConnectionPool
(facade)
PgWorkerPo
□□ □

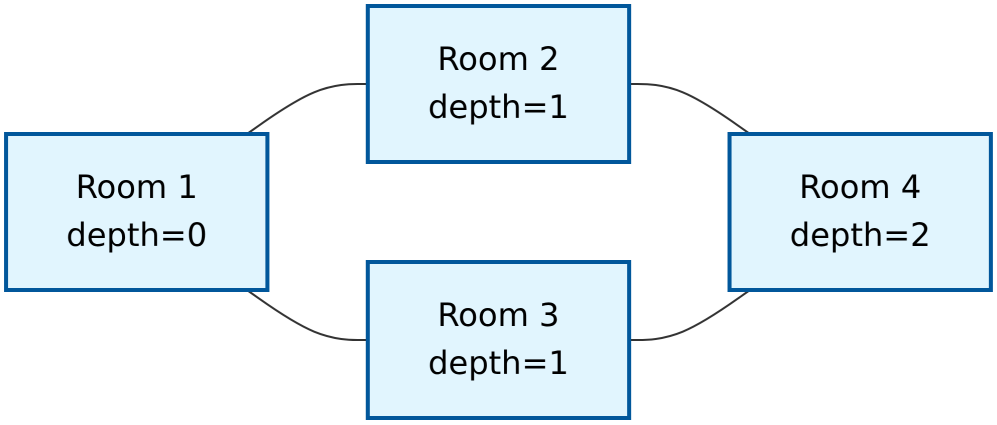
8: DB

5:

GRID	120×70
FILL	52% (+depth×2) ( )
WALL_THRESHOLD	15
ROOM_THRESHOLD	20
PASSAGE_RADIUS	1

7.4 Zone

Room . Room `connections` (`portal_index → target RoomId`) , .



9: Zone

:

1. Portal A → `C_PORTAL(portal_id)`
2. `old_room.Connections()[portal_id]` → Room ,
3. Room : + `depth = old_depth + 1`
4. Room (±2m) →

7.5 Zone

Room `shared_ptr` , `empty_since_` .

- `AddPlayer` `ClearEmpty()` —
- `RemovePlayer` `players_` `MarkEmpty()` —
- 10 `CleanupEmptyRooms()`
- : + 30 + Room
- Room `connections` Room

`AddPlayer` Room race condition .

7.6 DB

GameServer DB 6 `PgConnectionPool` **PostgreSQL** . · · · PostgreSQL , DB `libpq` `IoRing::Post` `IoWorker` . DB .

GameServer `DbService` (`PgDbService` / `SqliteDbService` / `MemoryDbService`) . `GAMESERVER_DB` .

PostgreSQL ( )	`./GameServer` . 6 `PgConnectionPool`
SQLite	`GAMESERVER_DB=sqlite ./GameServer` / . sqlite3 amalgamation vendored
In-Memory	,

7.6.1 PgDbService — DbService

`PgDbService` 6 `PgConnectionPool` . SQL , `PgResult` → (`CharInfo`/`ItemData`/`CurrencyData`) . DB , `IoRing::Post` `IoWorker` Room .

(: `Login` " → ", `CreateChar` " insert → seed") `pool->Execute()` . `IoWorker` , .

DDL bootstrap `PgConnection` . main `IoRing::Current()` DDL . bootstrap , .

7.7

(Dog Tag) . 3m `C_PICKUP` .

- `GroundItem` : ground_id, item_def_id, , , 15
- `S_GROUND_ITEM_SPAWN` / `S_GROUND_ITEM_DESPAWN`:
- `TryPickup()`: → `S_ITEM_ADD` + `S_GROUND_ITEM_DESPAWN`
- Room::OnTick
- ( )

7.8

`C_PORTAL` Room `OnPacket(`) .

5 `game_session.cpp --- OnPacket`

```
void GameSession::OnPacket(uint16_t msg_id, const std::byte* data, uint32_t len) {
    switch (static_cast<MsgId>(msg_id)) {
        case MsgId::C_PORTAL: handler::HandlePortal(*this, data, len); return;
        default: break;
    }
    if (state_ == SessionState::InRoom) HandleInRoomPacket(msg_id, data, len);
    else HandlePreRoomPacket(msg_id, data, len);
}
```

Room `HandlePacket` , switch .

8 — io_uring vs epoll

ServerCore(io_uring) epoll , I/O .

8.1

- : WSL2 Ubuntu 24.04, localhost, Intel i7 (8)
- : `game_bench` / `echo_bench` (`tools/run_bench.sh`)
- : Login → EnterGame → Move(20Hz) → Attack(2Hz, 4),

7:

	I/O	
io_uring	io_uring	IoWorker I/O Zone , ServerCore
Epoll	epoll	epoll , ServerCore

8.2 Echo

io_uring I/O . 64 , pipeline=1 .

8: Echo

	(echo/s)	p50 (s)	p99 (s)
10	218,498	30	157
50	305,183	152	367
100	326,913	293	594

100 **327K echo/s**, p50 **293 s** .

8.3 —

`C_MOVE` → `S_MOVE` (s) . .

9: p50 (, s)

		io_uring p50	Epoll p50
40	4	838	3,322
200	4	10,001	4,419,583
400	8	3,025	—

- **40** : io_uring , Epoll 3ms
- **200** : Epoll , io_uring 10ms
- **400** : io_uring 3ms , Epoll

8.4 Tail Latency — p99

10: Tail latency p99 (s)

		io_uring p99	Epoll p99
40	1	1,887	6,067
200	4	24,191	11,794,952

200 p99 , io_uring 24ms Epoll 11.8 .

8.5

`S_MOVE + S_ATTACK + S_DAMAGE` / .

11:

		io_uring (msg/s)	Epoll (msg/s)	
40	1	37,288	37,288	1.0×
200	4	950,656	523,846	**1.8×**

(40) , 200 Epoll .

8.6 Room — Room

Room Room 20 , O(N²) .

12: Room p50

	io_uring p50 (s)	Epoll p50 (s)
200 / 10 Room 4T×50C	846	897
400 / 20 Room 4T×100C	920	691

Room . Room , io_uring .

8.7

io_uring	Epoll
400 ms , Zone IO	, SQE 200

. 40 , 200 epoll p99 , io_uring 400 . Room , io_uring “ ” “ ” .